Project

Shiny app:

https://mireyas.shinyapps.io/Database_final_project/

Court Case Trends & DOA Risk Pattern Database

Mireya Sanchez

Introduction

The legal industry is under increasing pressure to operate more efficiently, with law firms and public legal offices alike seeking ways to reduce costs, allocate resources strategically, and anticipate case outcomes before investing significant time and capital. For firms that frequently interface with the District Attorney's Office (DAO), understanding prosecutorial patterns (such as how long cases typically take, how they are likely to resolve, and what factors influence those outcomes) can provide a meaningful competitive and operational advantage.

This project proposes the design and implementation of a relational database application that analyzes criminal case data sourced exclusively from the District Attorney's Office. The dataset encompasses all cases in which the DAO has taken action to prosecute, either by filing new criminal charges or by filing a Motion to Revoke Probation or Parole (MTR) and is updated on a weekly basis. By organizing this publicly available data into a normalized relational schema, the application supports structured querying and predictive analysis of prosecutorial activity.

The application is designed with two primary goals: first, to identify court case trends such as charge frequency, case volume fluctuations, and historical resolution patterns; and second, to detect DAO risk patterns that can inform law firm decision-making, which includes cases with elevated dismissal likelihood and prolonged processing timelines,. Ultimately, this system aims to help law firms predict case outcomes and estimated case length, enabling smarter resource allocation and more informed legal strategy.

By leveraging SQL as the backbone of this analytical system and deploying it as an interactive web application built in R Shiny, this project demonstrates how relational database design can transform raw prosecutorial records into actionable legal intelligence, all while relying exclusively on publicly available data and remaining within appropriate ethical boundaries.

Research Context

Court data analytics has become increasingly important in legal research, public administration, and judicial transparency. Scholars and policy analysts use court records to examine case backlogs, prosecutorial efficiency, and outcome disparities across case types (National Center for State Courts, 2023). Public access to court information is further supported by the Legal Information Institute and similar open-law initiatives, which emphasize transparency in judicial systems (Legal Information Institute, n.d.).

District Attorney's Offices are a particularly valuable source of prosecutorial data, as their records capture the full lifecycle of criminal cases from initial filing through final disposition. Federal courts provide electronic access to case metadata through systems such as PACER, while many state and county-level offices publish searchable online docket systems (Administrative Office of the U.S. Courts, n.d.). These datasets allow for the examination of:

- Filing frequency by charge type
- Case resolution times and processing timelines
- Distribution of dismissal, conviction, and probation outcomes
- Patterns in Motion to Revoke Probation or Parole (MTR) filings

However, raw court data is not always standardized, and differences in labeling, outcome descriptions, and reporting formats can limit analytical reliability. A structured relational database with controlled vocabulary for case type and outcome category can address this. It enables the kind of predictive analysis that allows law firms to anticipate case length, estimate dismissal likelihood, and allocate resources more strategically.

From a database design perspective, this project applies principles of relational modeling, entity normalization, and referential integrity as described in standard database systems literature (Silberschatz, Korth, & Sudarshan, 2020). By separating entities such as Incidents, Cases, Case Types, Disposition, Charges, and Custody, the design reduces redundancy and supports the scalable analytical queries that underpin the application's predictive capabilities.

Data Description

The dataset used in this project consists of criminal case records obtained from a public data source. Each record represents a case associated with a specific incident and includes information on arrest details, case filings, charges, and case outcomes.

Columns in the original dataset include:

· **incident number:** Police incident report number

· **court_number:** San Francisco Superior Court case number, unique to each defendant per incident

· **arrest_date:** The date of the arrest

· **list_of_filed_charges:** List of all charges (by California penal code designations) brought for the case; for MTRs, this list is made up of charges for the original case filed

· **filed_case_type:** Abbreviation for the most serious case type among the charges filed by SFDA for a given court number

· **crime_type:** Abbreviation for the most serious crime type among the charges filed by SFDA

· **dv_case:** Indicator for domestic violence cases

· **da_action_taken:** Action taken by District Attorney's office

· **case_status:** Current status of the case at its current point in the criminal process

· **data_as_of:** Timestamp when the data was last updated in the source system

· **data_loaded_at:** Timestamp when the data was last updated (in the data portal)

The original dataset is denormalized, with some fields containing multiple values (e.g., lists of charges). As part of the database design process, the data was transformed into a normalized relational schema consisting of separate tables for incidents, cases, charges, dispositions, and case types. This transformation improves data integrity, reduces redundancy, and enables more efficient querying and analysis within the database system.

Data Design

The database was designed using a relational model to organize and analyze criminal case data. The design process focused on transforming a denormalized dataset into a structured schema that supports efficient storage, querying, and analysis.As shown in Figure 1, the schema is centered on the 'cases' table, which links incident-level information, case classifications, charges, and case outcomes. To reduce redundancy and improve data integrity, the database was normalized into multiple related tables, including ‘incidents’, ‘case types’, ‘dispositions’, ‘charges’, and ‘custody’.

Incidents

The ‘incidents’ table stores information about police incidents. Each record represents a unique incident and includes attributes such as the incident number, arrest date, and domestic violence indicator. A surrogate key (`incident_id`) was introduced as the primary key to uniquely identify each incident and support relationships with other tables.

Cases

The ‘`cases’` table is the central entity in the database and represents individual court cases. Each case is associated with an incident and includes attributes such as filing date, case status, and crime type. A surrogate key (`case_id`) is used as the primary key, while foreign keys (`incident_id` and `case_type_id) es`tablish relationships with the ‘incidents’ and ‘case type’ tables.

Case Types

The ‘case types’ table functions as a lookup table that stores standardized classifications for each type of court case, such as felony or misdemeanor. Instead of repeatedly storing the same case type values in every record within the ‘cases’ table, each classification is stored once in the ‘case types’ table and referenced through a foreign key (case_type_id). This design reduces data redundancy, improves consistency across records, and supports grouping and analysis by case type.

Charges

The ’`charges’` table stores detailed charge information associated with each case. Because a single case can involve multiple charges, this table captures a one-to-many relationship between cases and charges. Each charge is stored as a separate row and linked to its corresponding case through a foreign key (case_id).

Dispositions

The ’`dispositions’` table records the outcome of each case, including the disposition date, code, and description. This table is linked to the ’`cases’` table through a foreign key (`case_id)`and allows for analysis of case outcomes and duration.

Custody

The custody table stores custody-related information for each case, including incarceration status, bond status, and bond amount. Each record is linked to the cases table through a foreign key ‘case_id’. Separating this information into its own table reduces redundancy, supports database normalization, and allows the dashboard to analyze custody risk and bond-related workload more effectively.

Normalization

The database schema was designed to satisfy Third Normal Form (3NF). First, the data was transformed into First Normal Form by removing repeating groups, such as the list of charges, into a separate '`charges`' table. Second, the use of single-column primary keys ensures that all non-key attributes depend on the entire key, satisfying Second Normal Form. The Third Normal Form was achieved by eliminating transitive dependencies through the use of lookup tables, such as '`case types`', which store descriptive attributes separately from the main case records. This normalization reduces redundancy, improves data integrity, and supports efficient querying.

Derived Attributes

Several derived attributes were created to support analytical functionality. The variable '`case_duration_days' was calculated as the difference between ‘disposition_date’ and ‘filing_date’ to measure the duration of each case. These derived variables enhance the analytical capabilities of the system while maintaining a normalized underlying schema.`

Surrogate Keys

Surrogate keys were created for major entities within the database, including ‘incident_id’, ‘case_id’, ‘case_type_id’, ‘disposition_id’, ‘charge_id’, and ‘custody_id’, to provide standardized and system-generated identifiers for each record. These keys ensure that cases and related information can be consistently referenced regardless of differences in reporting formats, naming conventions, or source data inconsistencies. By using surrogate keys instead of relying solely on raw case numbers or descriptive values, the database maintains relational integrity, improves consistency across tables, and supports more reliable joins, filtering, and analysis within the Shiny application.

Application Overview

A Shiny web application was developed to provide an interactive interface for analyzing and managing criminal case data stored within the relational database system. The application was designed to support legal operations and resource allocation by allowing users to explore case-level information, compare case outcomes, monitor custody and bond conditions, and evaluate workload trends. The dashboard integrates data from multiple normalized tables, including cases, incidents, dispositions, charges, case types, and custody information, into a user-friendly analytical platform.

The primary goal of the application is to help law firms and legal professionals identify high-priority cases, monitor operational workload, and better distribute legal resources based on case characteristics, outcomes, duration, and custody-related risk indicators.

Application Features

The application contains several interactive dashboard tabs that provide different analytical perspectives on the data.

Case Directory

The Case Directory serves as the primary operational view of the application. Users can search, filter, and review detailed case information, including case identifiers, crime types, filing dates, case status, case type classifications, custody conditions, and bond amounts. The search functionality was designed to support legal workflow continuity by allowing users to locate records using a 'case_id', 'incident_id', or 'disposition_id'.

This feature is especially useful because criminal matters may be referenced differently depending on the stage of the legal process. For example, a case may initially be assigned and tracked using an 'incident_id' during the incident reporting phase, later referenced by a 'case_id' once formal charges are filed, and ultimately associated with a 'disposition_id' after the case outcome has been recorded. By supporting searches across all three identifiers, the application allows users to locate and track the same matter regardless of which phase-specific identifier is available. This improves operational efficiency and helps ensure continuity in case management across different stages of legal processing.

Case Portfolio Summary

The Demographic Summary tab provides visual summaries of the firm's overall caseload using interactive charts that display case type distributions, case outcomes, and crime category patterns. This section was designed to help users quickly understand the composition of the legal workload and identify trends that may influence staffing, attorney specialization, and operational planning.

The Case Type Summary chart compares felony and misdemeanor cases within the dataset, allowing users to evaluate how much of the caseload involves potentially higher-complexity matters. For example, the dashboard currently shows that misdemeanor cases make up the majority of the portfolio, while felony cases represent a smaller but potentially more resource-intensive portion of the workload.

The Case Status Summary chart displays the distribution of legal outcomes, including convictions, dismissals, diversions, and plea-related resolutions. This visualization helps users understand how cases typically progress through the legal system and identify operational trends that may affect attorney preparation, negotiation strategies, or case management priorities.

The Crime Type Summary chart provides a breakdown of crime categories represented within the dataset, including offenses such as DUI, burglary, assault, kidnapping, and narcotics-related charges. This feature allows users to identify which offense categories occur most frequently and evaluate how different crime types may contribute to overall legal workload demands. Together, these visualizations provide a broad analytical overview of the firm's caseload and support more informed resource allocation and operational decision-making.

Case Status Comparison

The Case Status Comparison tab analyzes how legal outcomes differ between felony and misdemeanor cases by comparing the frequency of each case status category across case types. The visualization was designed to help users evaluate outcome trends and understand how different categories of criminal cases progress through the legal system.

The stacked bar chart compares outcomes such as convictions, dismissals, diversions, and plea-related resolutions for both felony and misdemeanor cases. By displaying the number of cases within each status category, the dashboard allows users to quickly identify which types of cases are more likely to result in specific legal outcomes.

For example, the visualization shows that convictions represent the largest outcome category for both felony and misdemeanor cases, while dismissals and diversions occur less frequently. The chart also demonstrates that misdemeanor cases make up the majority of outcomes across nearly every status category, reflecting the larger proportion of misdemeanor cases within the dataset. Felony cases, although fewer in number, still contribute significantly to conviction and dismissal outcomes, suggesting that these matters may require greater legal preparation and resource allocation despite their smaller volume.

This tab supports operational decision-making by helping legal professionals identify trends in case resolution patterns, evaluate workload distribution between case categories, and anticipate which types of matters may require additional attorney attention, negotiation efforts, or court preparation.

Case Duration Analysis

The Case Duration tab evaluates how cases progress over time by comparing filing activity and disposition activity across multiple years. This section was designed to help users identify workload trends, monitor case processing patterns, and evaluate how long cases remain active within the legal system. Understanding case duration is important for law firms because longer-running cases often require increased attorney time, court preparation, and operational resources.

The line chart titled *"Cases Filed and Disposed by Year"* compares the number of cases filed each year against the number of cases reaching disposition. The visualization helps users evaluate whether case resolutions are keeping pace with incoming workload. For example, the chart shows a high concentration of filings and dispositions during earlier years in the dataset, followed by lower activity levels in later years. This type of trend analysis can help legal professionals identify periods of heavier operational demand and evaluate how efficiently cases move through the legal process.

The tab also includes a summary table displaying the total number of felony and misdemeanor cases within the dataset. Once fully populated, these metrics would allow users to compare how long felony and misdemeanor cases typically remain active and identify categories of cases that may require additional legal resources or extended attorney involvement.

Overall, this tab supports operational planning by helping users monitor case processing timelines, evaluate workload trends over time, and identify opportunities to improve legal workflow efficiency.

Bond Comparison and Custody Analysis

The Bond Comparison tab focuses on custody-related conditions and financial bond information associated with criminal cases. This section was designed to help legal professionals identify higher-risk cases, prioritize attorney attention, and evaluate how custody status varies across different case categories and crime types. Because custody and bond conditions can significantly impact legal urgency and client needs, this tab supports operational decision-making and resource allocation within a law firm's environment.

The dashboard includes interactive filters that allow users to narrow results by case type and crime category. Users can select specific offense categories or focus only on felony or misdemeanor cases to evaluate how custody conditions differ across the portfolio. This filtering functionality allows attorneys and legal staff to isolate groups of cases that may require specialized handling or increased legal resources.

The *"Custody by Case Type"* visualization summarizes the number of incarcerated clients, clients released on bond, and dismissed cases across selected case categories. The chart demonstrates that incarceration and dismissal statuses make up a significant portion of the filtered dataset, while fewer individuals are currently out on bond. This type of analysis can help law firms identify how many active matters may require urgent hearings, custody reviews, or additional client communication.

The *"Custody by Crime Type"* visualization extends this analysis by comparing custody-related outcomes across individual crime categories. For example, users can filter the dashboard to examine custody conditions for offenses such as burglary, assault, DUI, narcotics, or kidnapping. This allows legal professionals to evaluate which offense categories are more frequently associated with incarceration or higher legal risk.

The dashboard also includes summary tables displaying the average bond amount and total number of cases within the selected filters. In the current filtered view, the average bond amount is approximately $2,166 across 30 cases. Bond analysis provides additional operational insight because higher bond amounts may indicate cases involving increased legal complexity, higher client risk, or greater urgency for legal intervention.

Overall, this tab supports legal resource management by helping users monitor custody conditions, evaluate financial bond patterns, identify high-priority cases, and better distribute attorney workload based on case severity and client custody status.

Conclusion

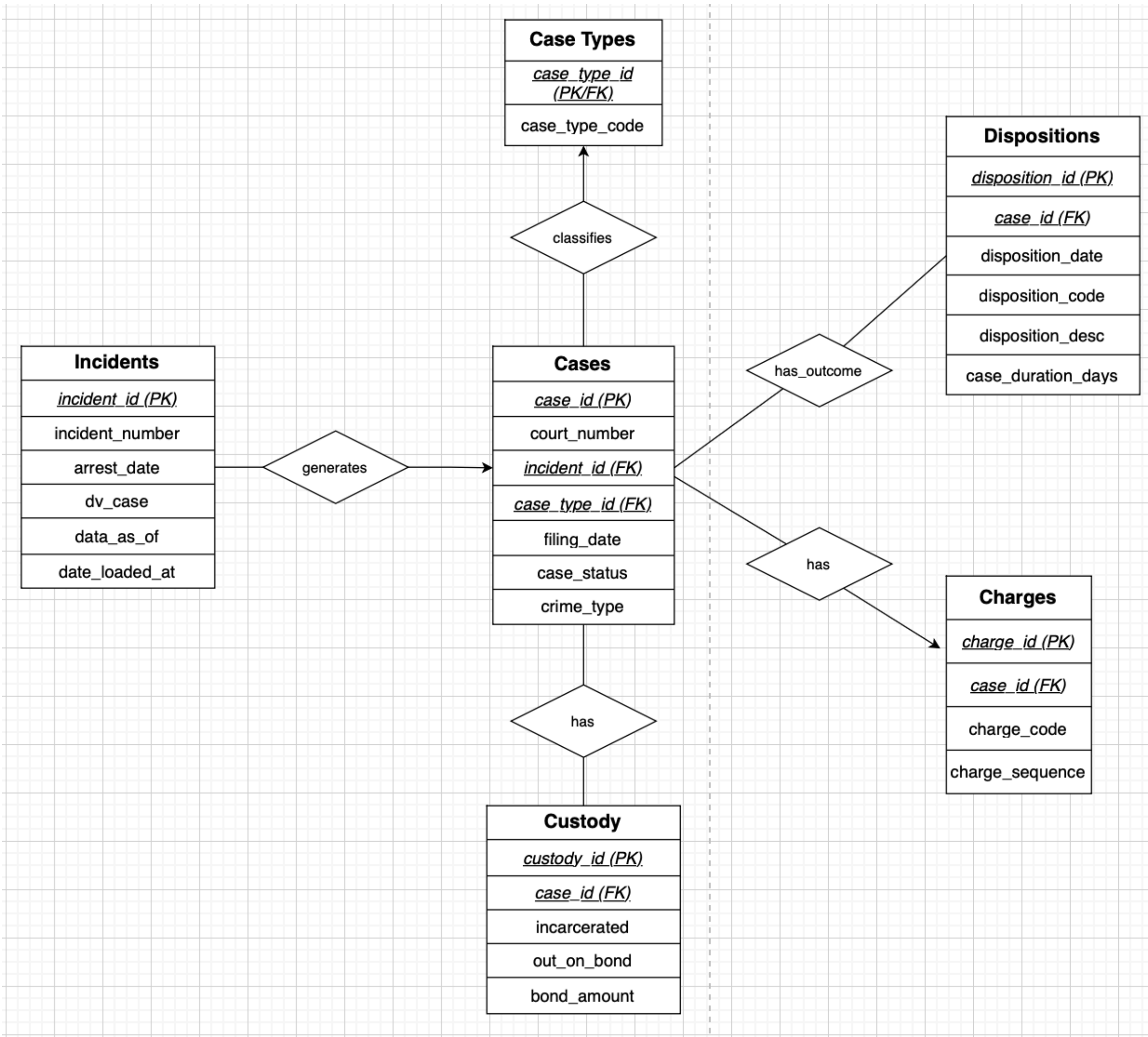
This project illustrates the value of applying relational database design principles to publicly available prosecutorial data to support structured analysis of criminal case activity. By transforming a denormalized dataset sourced from the District Attorney's Office into a normalized relational schema satisfying Third Normal Form, the database achieves improved data integrity, reduced redundancy, and efficient querying across cases, charges, dispositions, custody records, and case type classifications.

The R Shiny web application built on this database provides an interactive platform for legal professionals to explore case trends, monitor workload distribution, and evaluate outcome patterns across felony and misdemeanor categories. Each dashboard feature was designed to support a specific operational purpose: helping law firms identify high-priority cases, anticipate case trajectories, and allocate attorney resources more strategically based on case characteristics, outcomes, and custody-related risk indicators.

More broadly, this project demonstrates that structured relational modeling and schema normalization can transform raw prosecutorial records into a reliable foundation for legal analysis. As court systems continue to expand public access to case data, applications of this kind offer law firms a meaningful operational advantage. Future development could extend the system's analytical capabilities through integration with additional jurisdictional datasets, automated data pipeline updates aligned with the DAO's weekly refresh schedule, or expanded predictive modeling based on historical disposition and case duration patterns. Ultimately, this system confirms that publicly available court data, when properly organized and analyzed, can support more informed legal decision-making while remaining within appropriate ethical boundaries

Appendix

Figure 1


Case Types
case_type_id (PK/FK)
case_type_code
classifies
Dispositions
disposition_id (PK)
case_id (FK)
disposition_date
disposition_code
disposition_desc
case_duration_days
Incidents
incident_id (PK)
incident_number
arrest_date
dv_case
data_as_of
date_loaded_at
generates
Cases
case_id (PK)
court_number
incident_id (FK)
case_type_id (FK)
filing_date
case_status
crime_type
has_outcome
has
Charges
charge_id (PK)
case_id (FK)
charge_code
charge_sequence
has
Custody
custody_id (PK)
case_id (FK)
incarcerated
out_on_bond
bond_amount